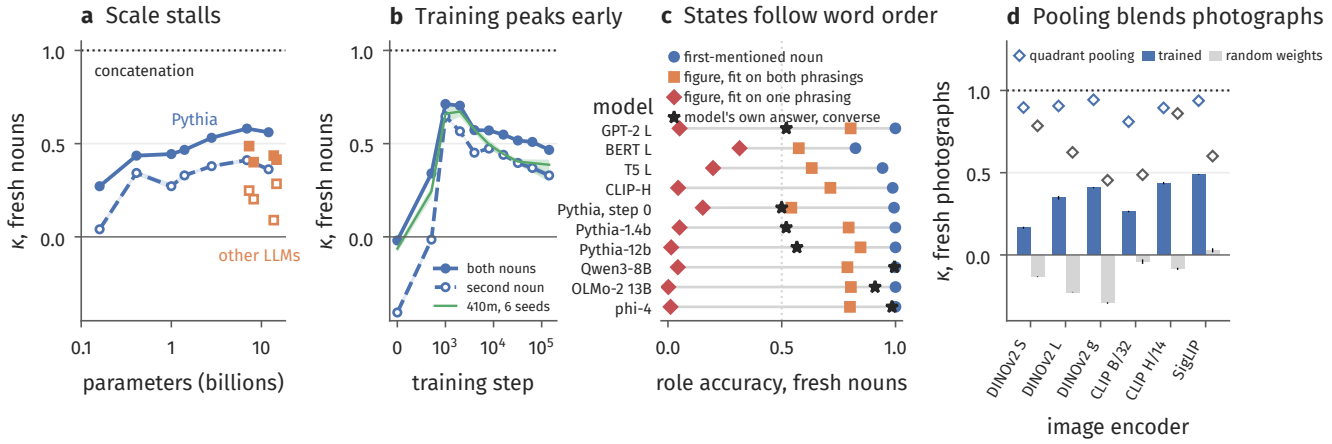

a Scale stalls
1.0
0.5
0.0
κ, fresh nouns
concatenation
Pythia
other LLMs
0.1
1
10
parameters (billions)
b Training peaks early
1.0
0.5
0.0
κ, fresh nouns
both nouns
second noun
410m, 6 seeds
0
10^3
10^4
10^5
training step
c States follow word order
first-mentioned noun
figure, fit on both phrasings
figure, fit on one phrasing
model's own answer, converse
model
GPT-2 L
BERT L
T5 L
CLIP-H
Pythia, step 0
Pythia-1.4b
Pythia-12b
Qwen3-8B
OLMo-2 13B
phi-4
0.0
0.5
1.0
role accuracy, fresh nouns
d Pooling blends photographs
quadrant pooling
trained
random weights
1.0
0.5
0.0
κ, fresh photographs
DINOv2 S
DINOv2 L
DINOv2 g
CLIP B/32
CLIP H/14
SigLIP
image encoder